

3D Volume Image Analysis for Biologists

Introduction to artificial intelligence for object segmentation

Slide curtesy : DBI IACF PhD Course 2026

NUS National University of Singapore
Microscopy Cluster
Yong Loo Lin School of Medicine



Content

- Introduction to ML/DL
- Use and retrain Empanada
- Use and retrain Cellpose

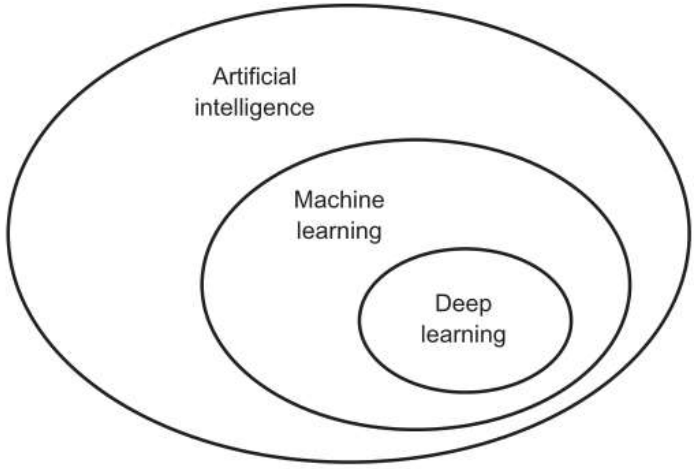
General introduction

AI: The effort to automate intellectual tasks normally performed by humans

François Chollet, *Deep Learning with Python* (2018)

Machine learning (60' … 85' … 10' - present):
A field of study that gives computers the ability to learn without being explicitly programmed.

Arthur L. Samuel (1959)



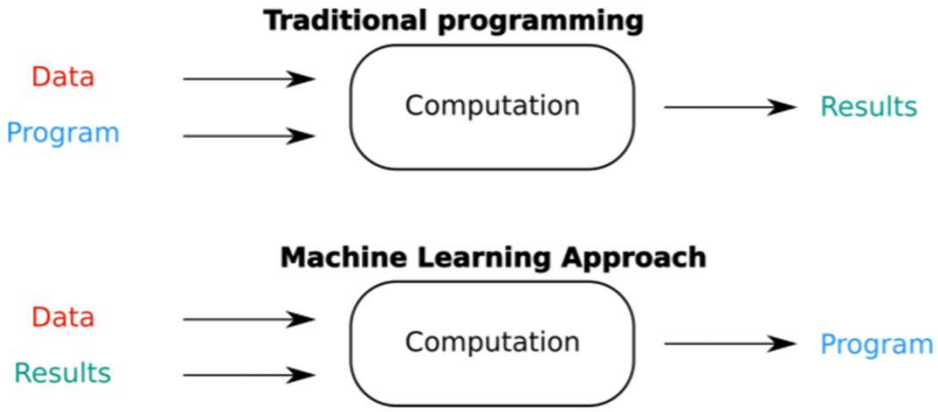
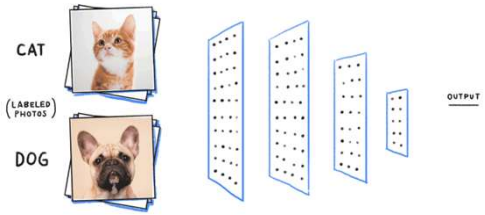


Image related problems

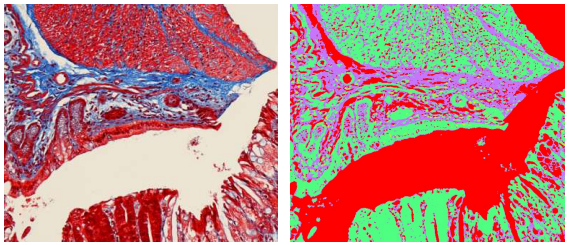
Classification

Image classification



1 class / image

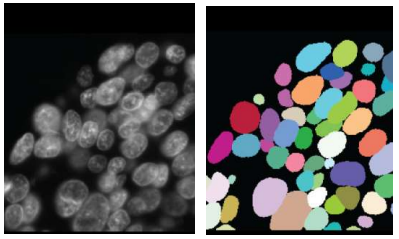
Semantic classification



1 class / pixel

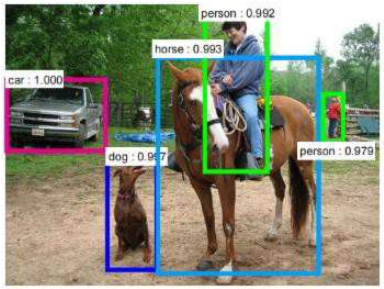
Segmentation / Tracking

Instance segmentation



Independent objects

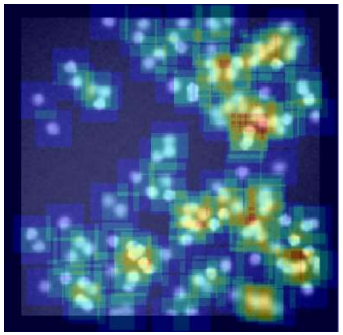
Object detection



Bounding boxes + classes

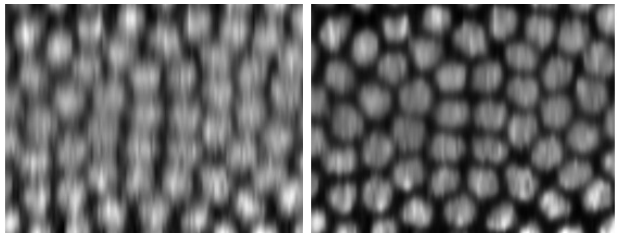
Regression + Classification

Object counting



Number

Image restoration

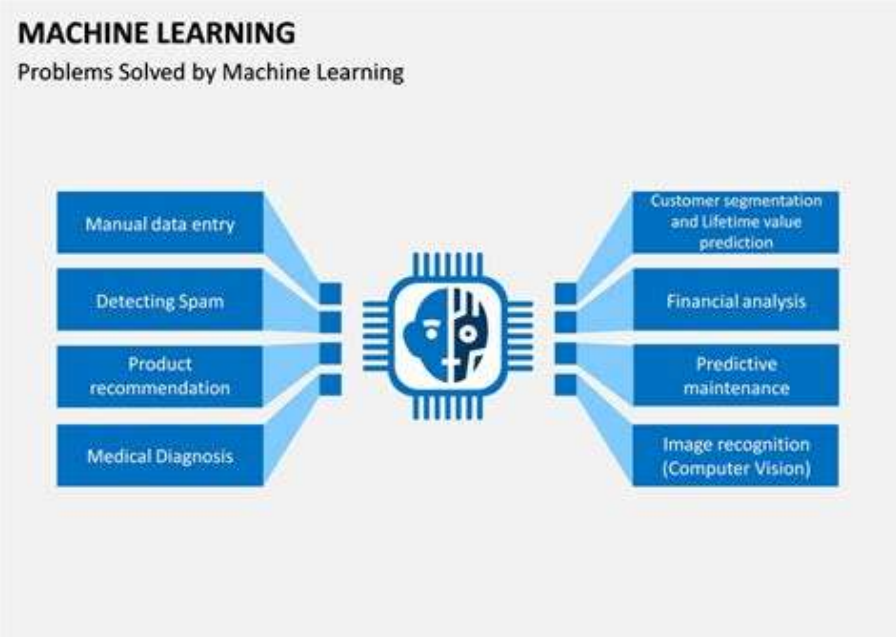


Image

Regression

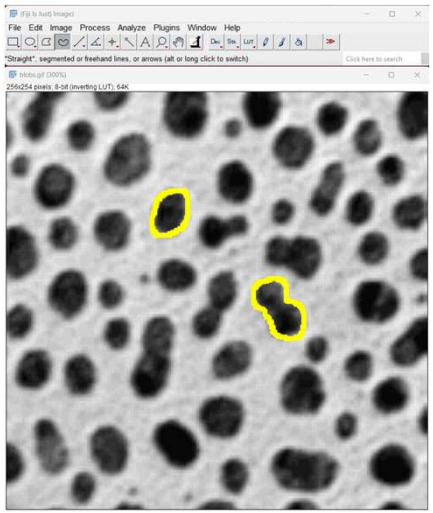
Why use machine learning

- Re-usability (on same problem)
- Genericity (on different problems)
- No human biais
- Accuracy
  - Near-human: translation, handwriting transcription, speech recognition, image classification
  - Super-human: playing chess & Go, lymph nodes metastases detection

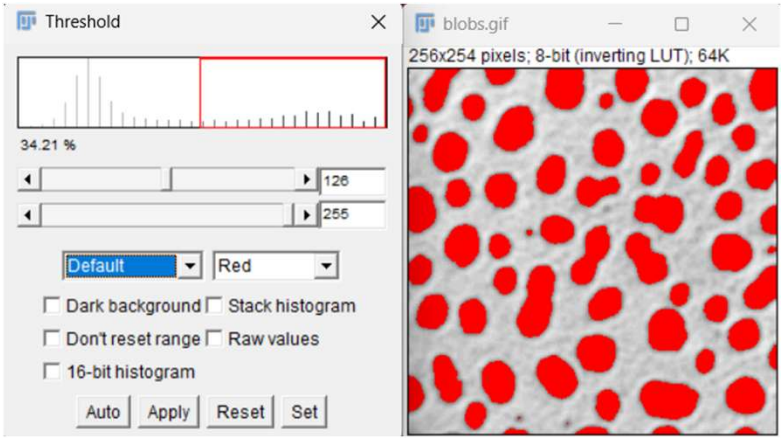


Use case scenario

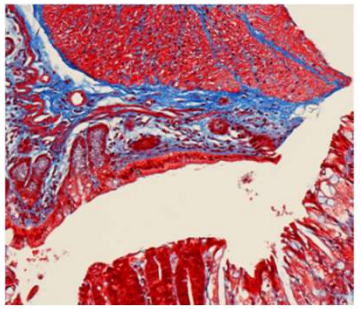
Manual Drawing

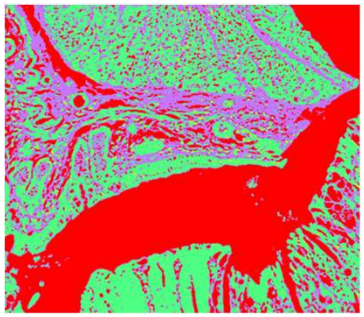


Manual/Automatic Thresholding



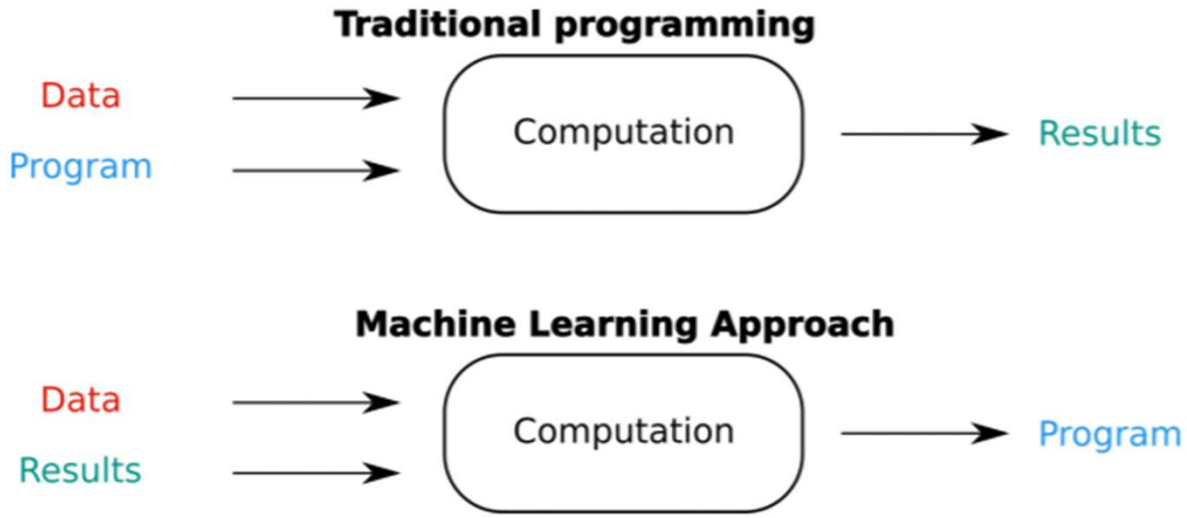
But what if you want to assign every pixel to a class? What might happen if you segment each signal one-by-one?





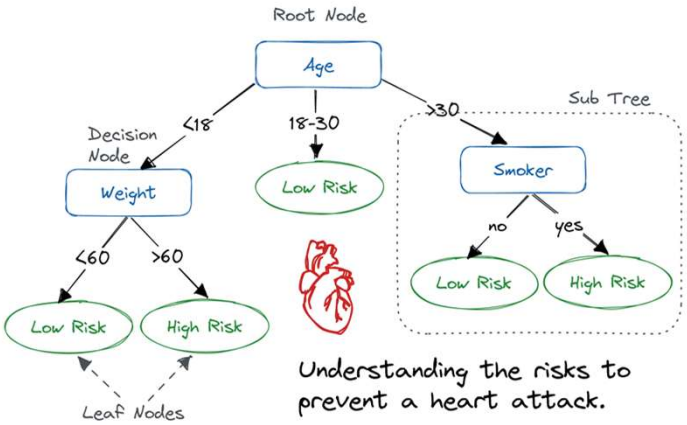
ML segmentation methods can be helpful here!

Machine learning versus Classical approach



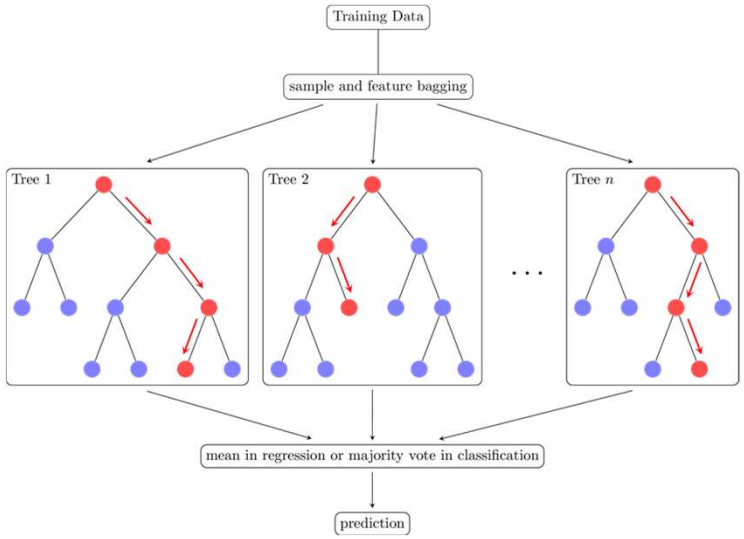
https://www.kaggle.com/getting-started/150361

Random Forests (Decision Trees)



Decision Tree: Create model that predicts outcome by learning rules to split features values

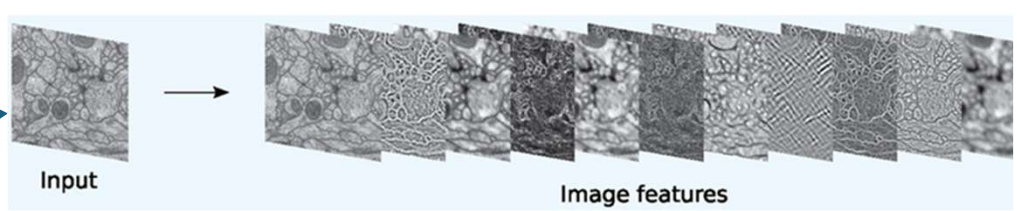
- Can result in complex, over-fitted trees
- Prone to find branch-specific solutions
  → Tree may not be optimal

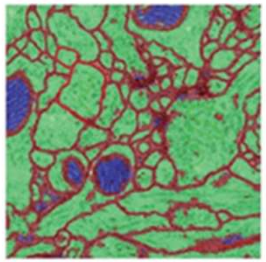


Random Forest: Ensemble of decision trees trained on random subsets of the training data.

The mean or majority vote is used predict the target value

Feature Based Random Forest Pixel Classifier





Segmentation

Data

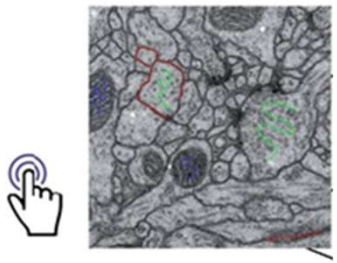
Answer

Machine Learning

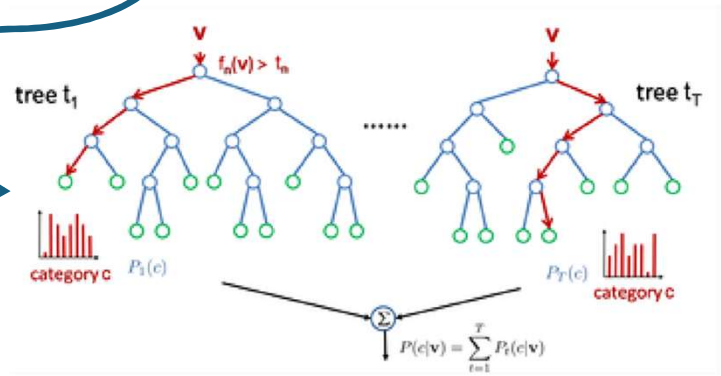


Rules (Model)

Prediction



Input labeling



Open Source Software for Pixel Classifier

- Standalone



ilastik

Perspective | Published: 30 September 2019

ilastik: interactive machine learning for (bio)image analysis

Stuart Berg, Dominik Kutra, Thorben Kroeger, Christoph N. Straehle, Bernhard X. Kausler, Carsten Haubold, Martin Schiegg, Janez Ales, Thorsten Beier, Markus Rudy, Kemal Eren, Jaime I Cervantes, Buote Xu, Fynn Beuttenmueller, Adrian Wolny, Chong Zhang, Ullrich Koethe, Fred A. Hamprecht & Anna Kreshuk

Nature Methods **16**, 1226–1232 (2019) | Cite this article

- Fiji/ImageJ

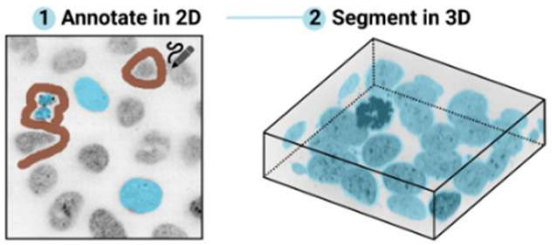


WEKA
The University of Waikato

(or labkit)

- Napari

Convpaint

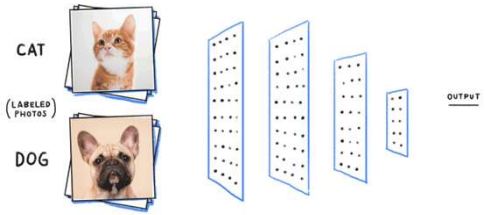


(or APOC: Accelerated Pixel Object Classifer)

Image related problems

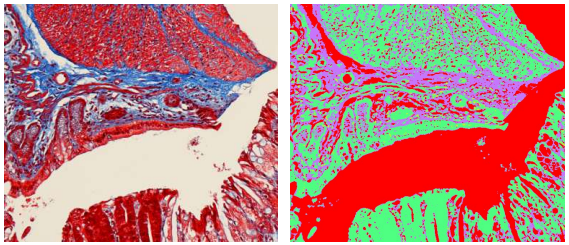
Classification

Image classification



1 class / image

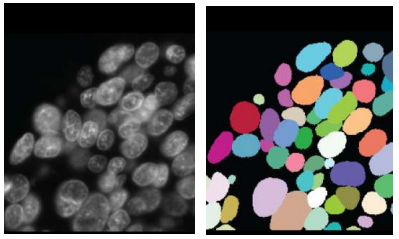
Semantic classification



1 class / pixel

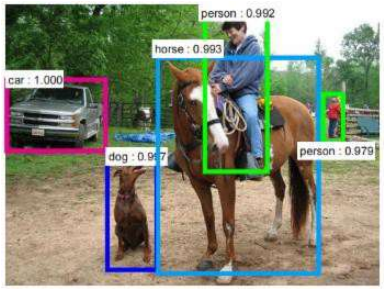
Segmentation / Tracking

Instance segmentation



Independent objects

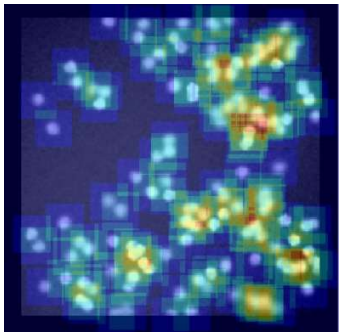
Object detection



Bounding boxes + classes

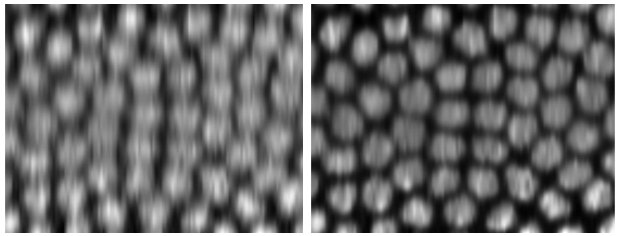
Regression + Classification

Object counting



Number

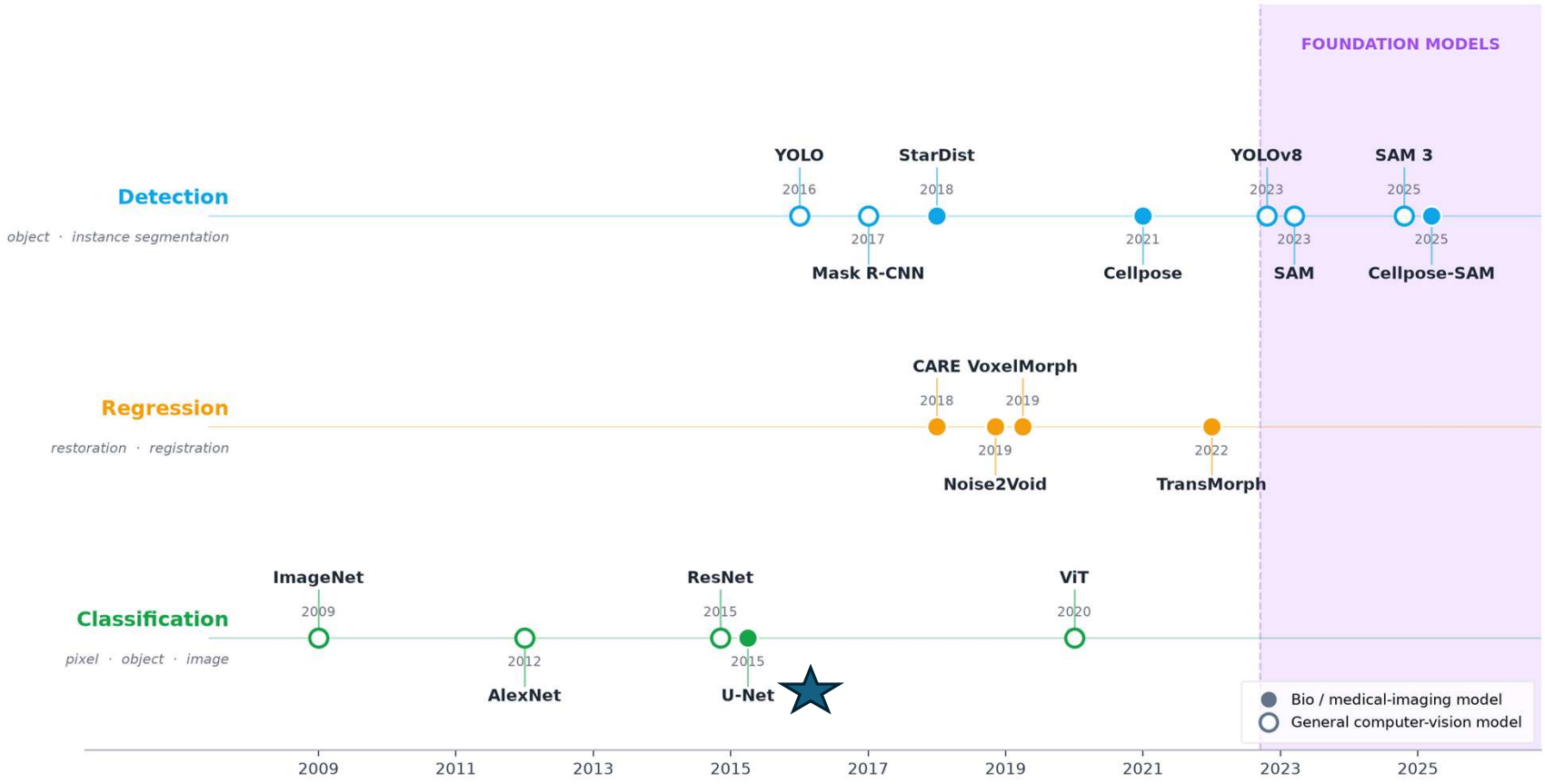
Image restoration



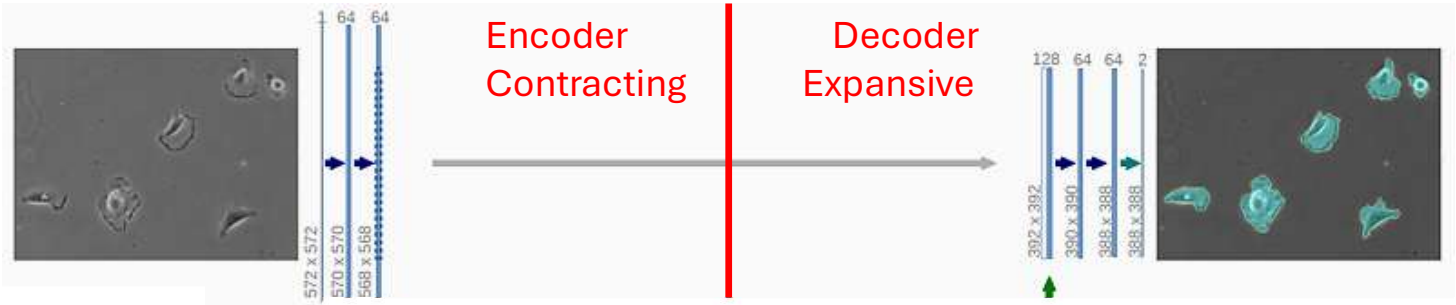
Image

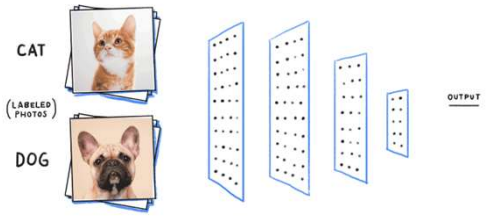
Regression

10+ years of rapid developments



CNN for Pixel Segmentation: U-NET





26 convolutional l
8.6M parameters



Olaf Ronneberger

FOLLOW

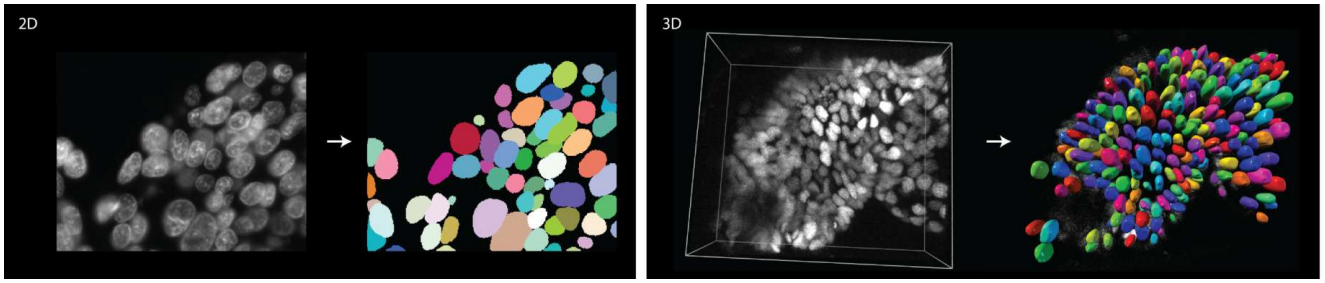
apl. Prof. for Computer Science, University of Freiburg, Germany
Verified email at cs.uni-freiburg.de - Homepage

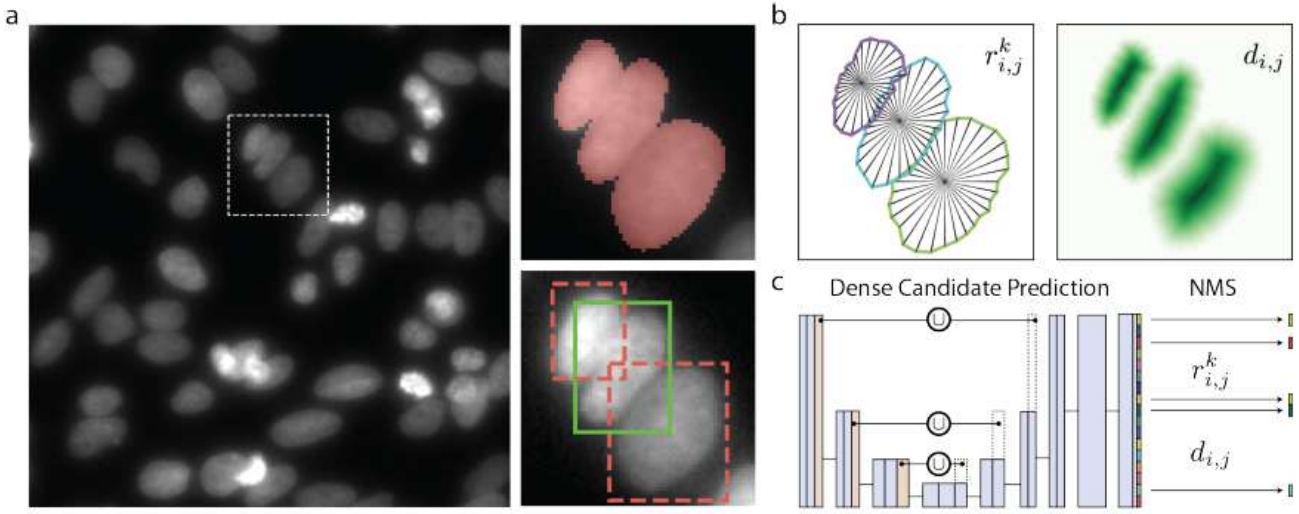
Image Analysis Computer Vision Pattern Recognition Biomedical Applications

TITLE	CITED BY	YEAR
U-net: Convolutional networks for biomedical image segmentation O Ronneberger, P Fischer, T Brox International Conference on Medical image computing and computer-assisted …	151662	2015

35 citations/day

Nuclei segmentation: Stardist





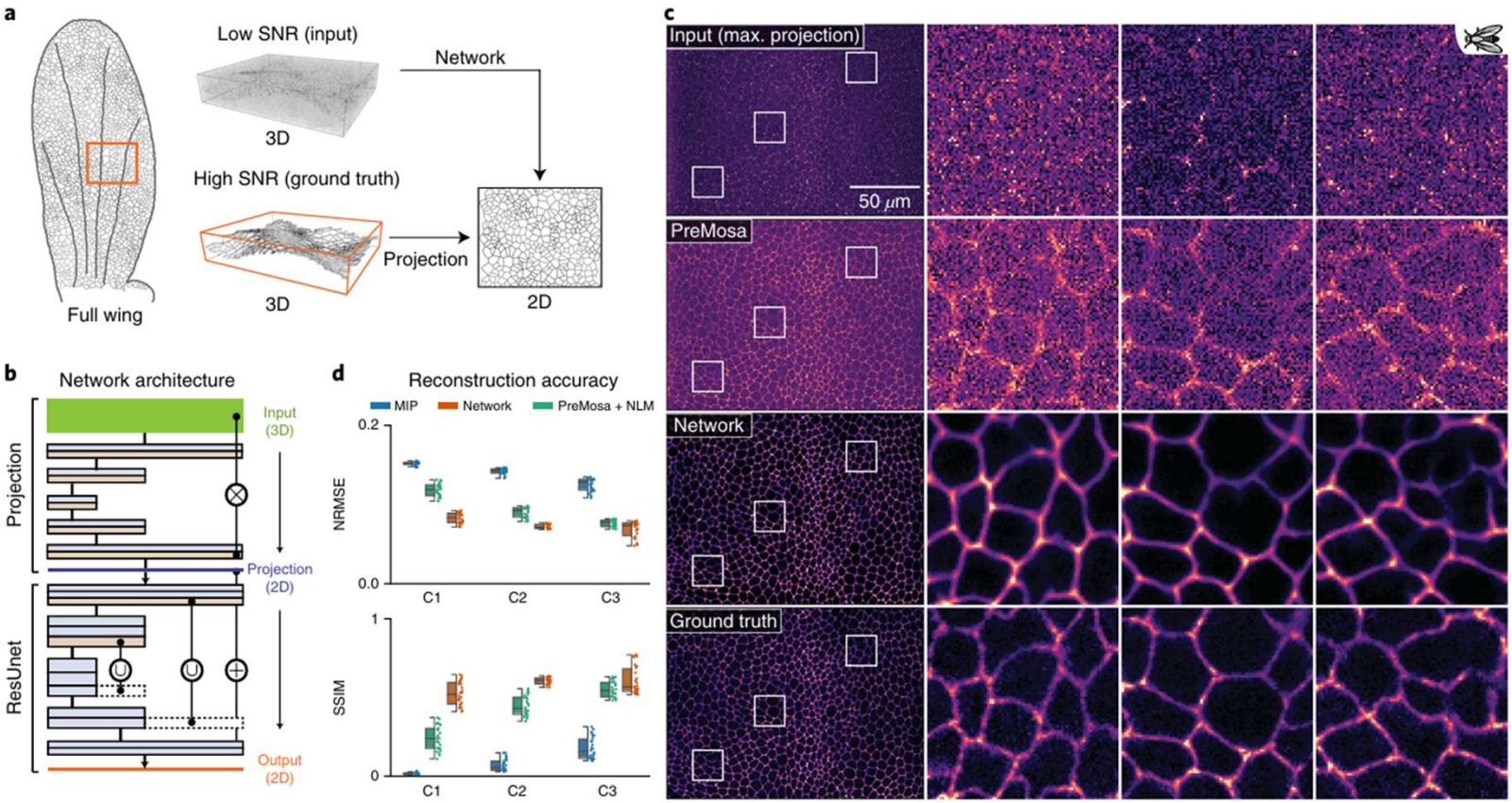
Availability:

- ImageJ plugin (Prediction 2D only)
- Python Notebooks (2D/3D) from both ZeroCostDL4Mic and CSBDeep
- Napari plugin (2D and 3D)

https://github.com/stardist/stardist
https://imagej.net/StarDist

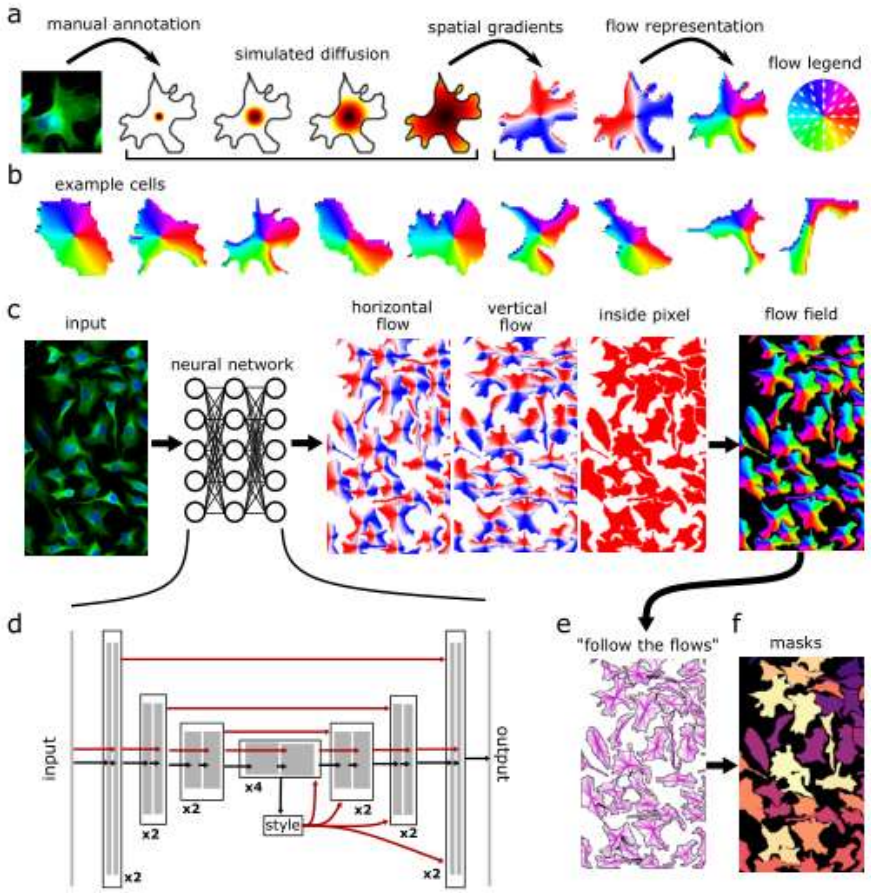
Schmidt et. Al, MICCAI, 2018

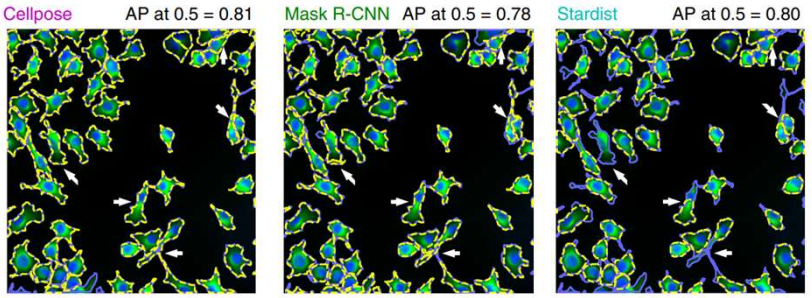
Content Aware Image Restoration



Weigert, M., Schmidt, U., Boothe, T. *et al*. *Nat Methods* **15** (2018)

Cellpose





Article | Published: 14 December 2020

Cellpose: a generalist algorithm for cellular segmentation

Carsen Stringer, Tim Wang, Michalis Michaelos & Marius Pachitariu

Nature Methods **18**, 100–106 (2021) | Cite this article

Article | Open access | Published: 07 November 2022

Cellpose 2.0: how to train your own model

Marius Pachitariu & Carsen Stringer

Nature Methods **19**, 1634–1641 (2022) | Cite this article

Article | Open access | Published: 12 February 2025

Cellpose3: one-click image restoration for improved cellular segmentation

Carsen Stringer & Marius Pachitariu

Nature Methods **22**, 592–599 (2025) | Cite this article

New Results

Follow this preprint

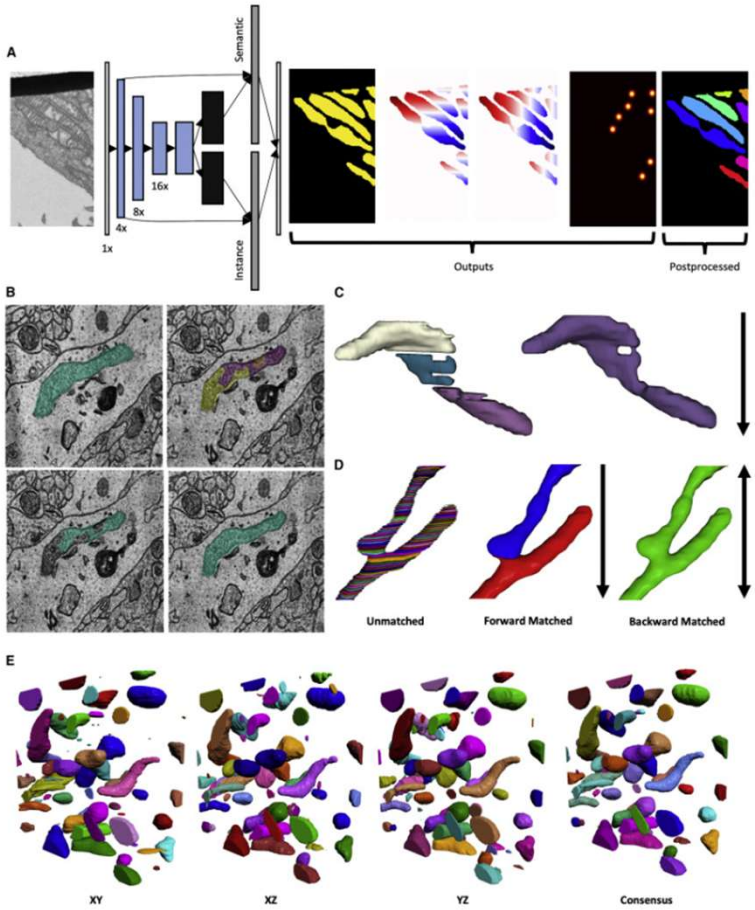
Cellpose-SAM: superhuman generalization for cellular segmentation

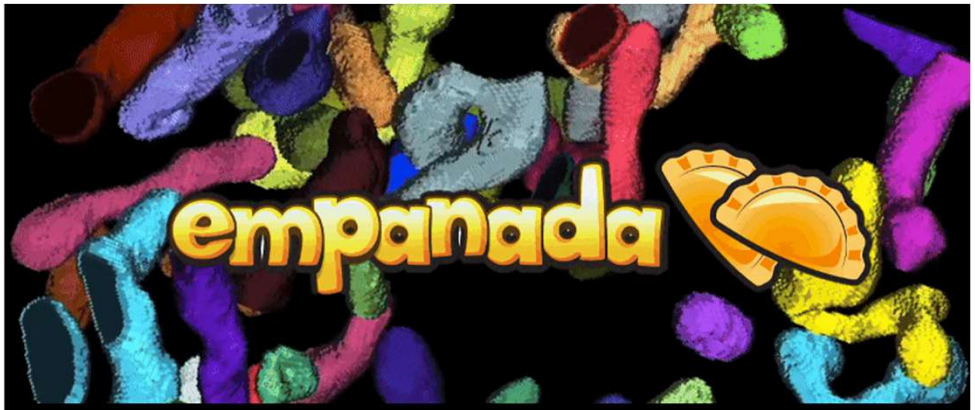
Marius Pachitariu, Michael Rariden, Carsen Stringer

doi: https://doi.org/10.1101/2025.04.28.651001

This article is a preprint and has not been certified by peer review [what does this mean?].

Empanada





Empanada (EM Panoptic Any Dimension Annotation) is a tool for panoptic segmentation of organelles in electron microscopy (EM) images in 2D and 3D. Panoptic segmentation combines both instance and semantic segmentation enabling a holistic approach to image annotation with a single deep learning model.

› Cell Syst. 2023 Jan 18;14(1):58-71.e5. doi: 10.1016/j.cels.2022.12.006.

Instance segmentation of mitochondria in electron microscopy images with a generalist deep learning model trained on a diverse dataset

Ryan Conrad [1], Kedar Narayan [2]

DISCLAIMER

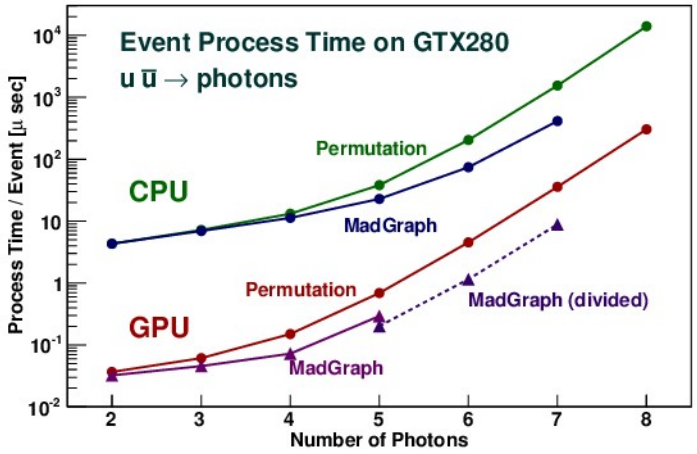
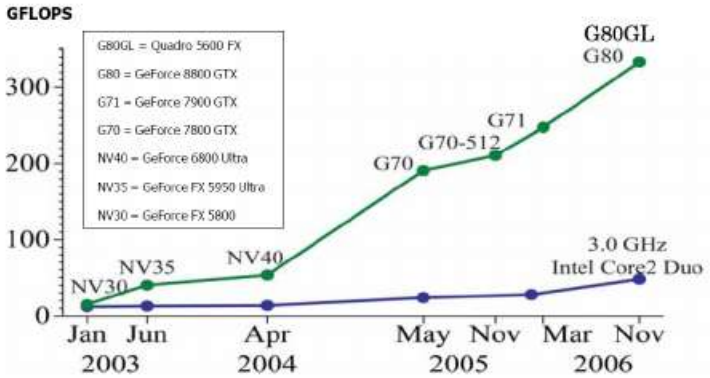


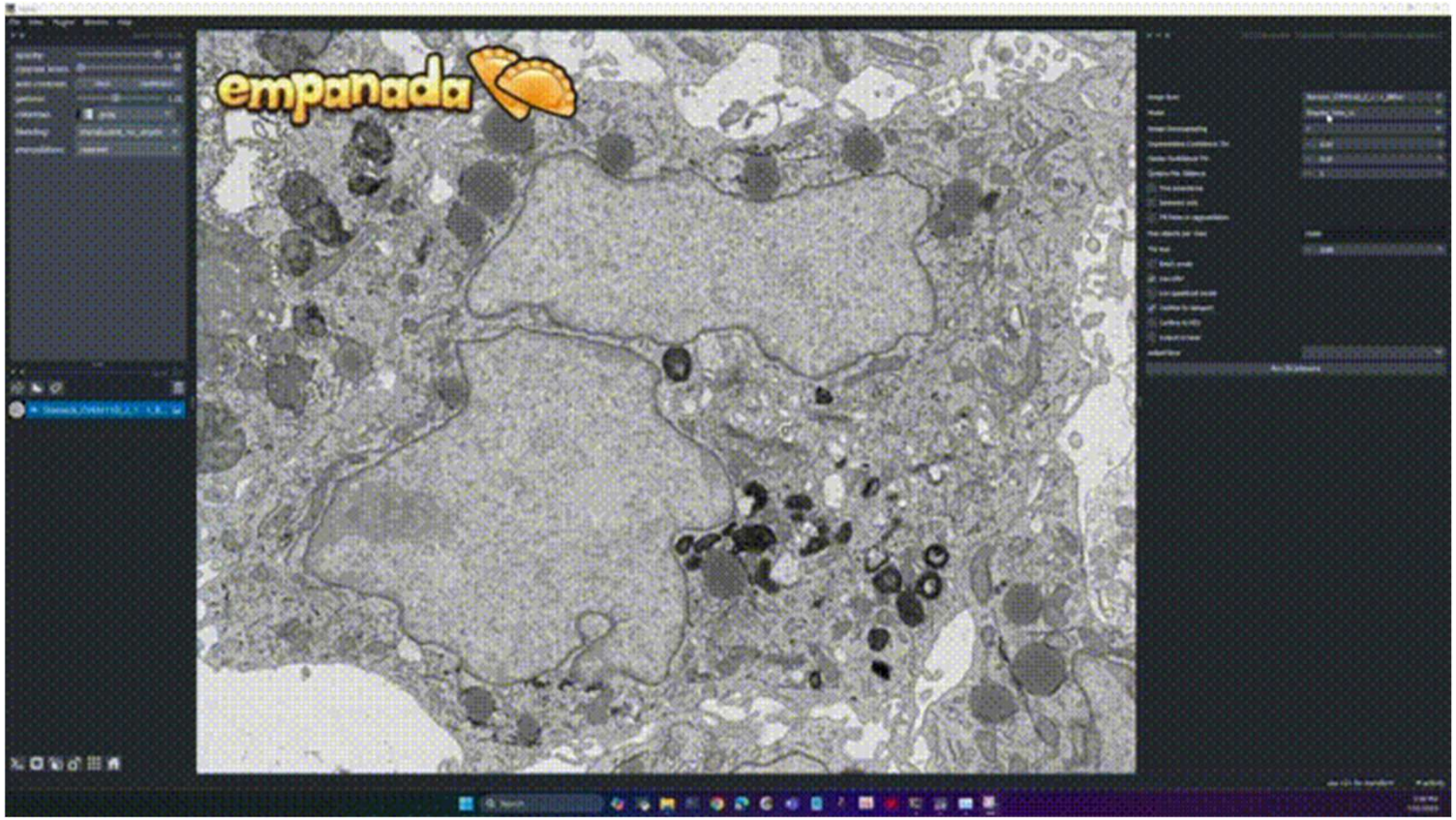
Image size(pixel)	CPU time(ms/image)	GPU time (ms/image)	Speed up
380*240	18	1	18.00
720*480	68	2	34.00
1280*720	206	5	41.20
1920*1280	601	14	42.93
2880*1920	1173	23	51.00
3840*2560	2610	50	52.20
4080*3720	4437	79	56.16
5616*3744	6162	109	56.53
6550*4080	8129	143	56.85
7120*5010	10722	185	57.96
8520*7120	16895	291	58.06



Hagiwara & al, EPJ 2010

Shams & al, IEEE 2008

Zhang & al, IOP 2014


empanada



Empanada (EM Panoptic Any Dimension Annotation) is a tool for panoptic segmentation of organelles in electron microscopy (EM) images in 2D and 3D. Panoptic segmentation combines both instance and semantic segmentation enabling a holistic approach to image annotation with a single deep learning model.

Getting Started

Step-by-step **installation instructions** for the latest version for new or returning users

Getting Started

Modules

Detailed descriptions of parameters, outputs and demo videos of all the modules included in empanada-napari

Modules

Tutorials

Indepth tutorials on how to finetune, train and run inference through this plugin

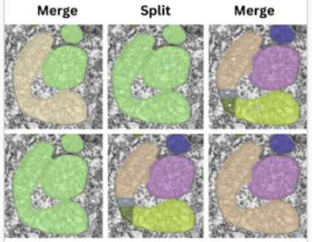
Tutorials

Share your model

Share your trained model with the us and the vEM community

Share your model

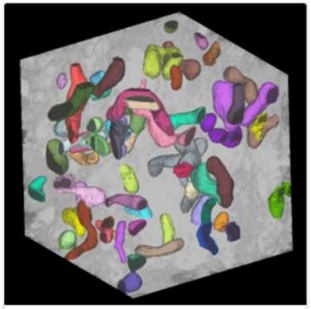
Tutorials



Step by step guide for 2D image inference.

Learn how to run 2D inference in empanada-napari with this step-by-step guide.

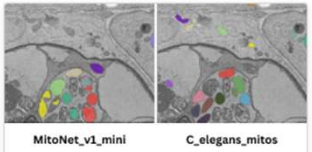
2D Inference Tutorial



Step by step guide for volumetric inference.

Apply similar techniques from the 2D inference tutorial for running inference on volumetric data.

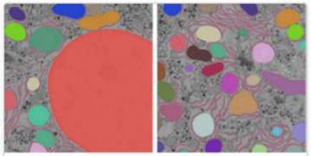
3D Inference Tutorial



Finetune an existing model

Follow the finetuning an existing model's step-by-step guide to help you learn how to tailor MitoNet_mini for a specific dataset's needs.

Finetuning Tutorial



Train a panoptic segmentation model

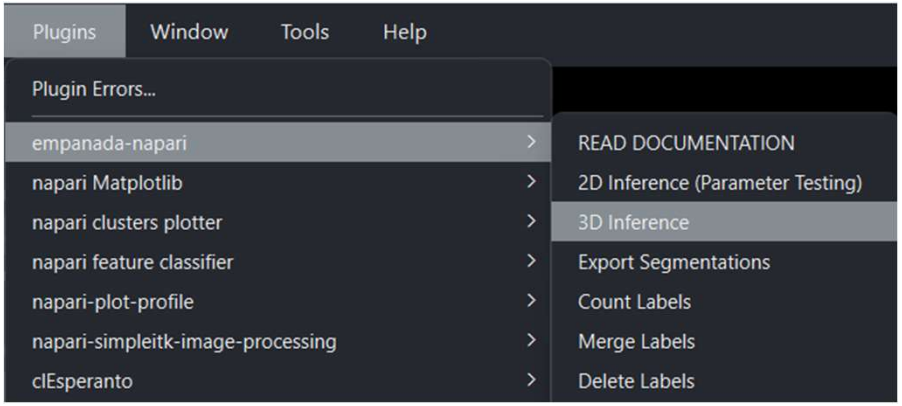
Use the training a panoptic segmentation model tutorial to go through the steps of creating training patches, adding panoptic labels, and other useful steps to fully train a model in empanada-napari.

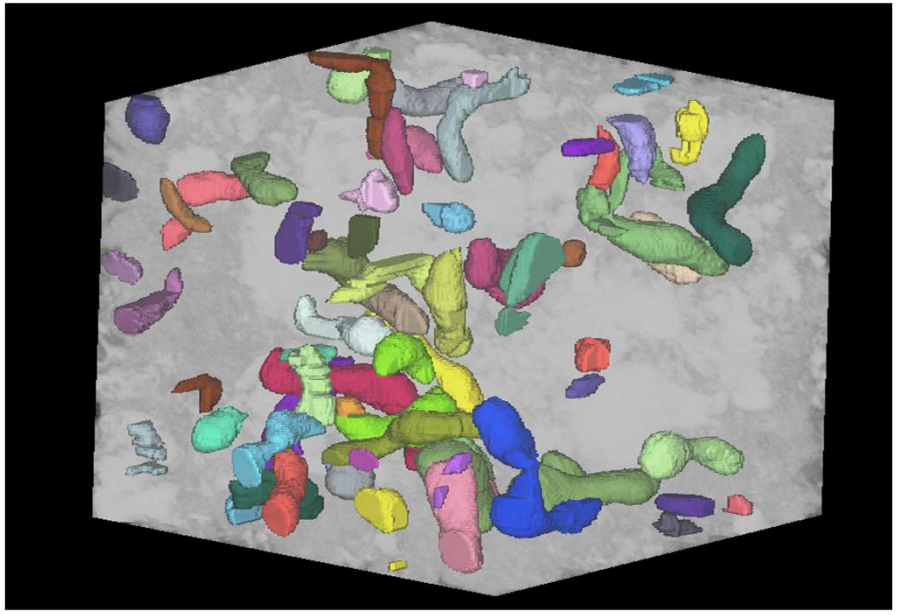
Training Tutorial

https://empanada.readthedocs.io/en/latest/

3D Segmentation

Open hela_cell_em.tif
Run 3D inference
Plugins > Empanada > 3D Inference
Change model to MitoNet_v1_mini
Run using all default parameters
(Check the terminal/prompt for progress)





Retrain Empanada

Open all images in Training > Empanada (use napari-builtin loader)
Select All, Right click and “Merge to Stack”
Create points layer (change point size to 70)
Put 1-2 points per slice around mitochondria (minimum 16 points)
Check model information

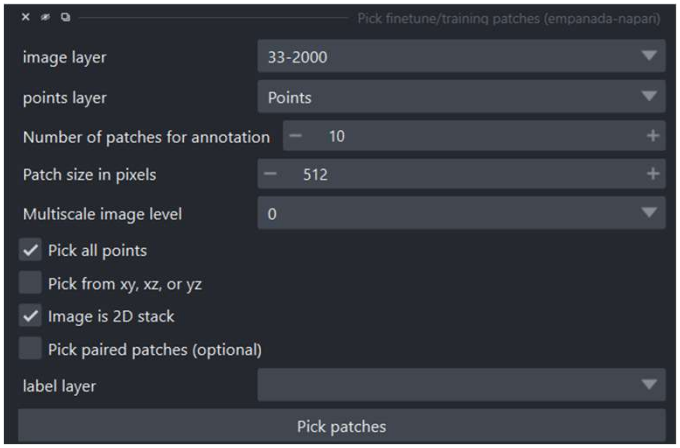
Plugins > Empanada > Get model info

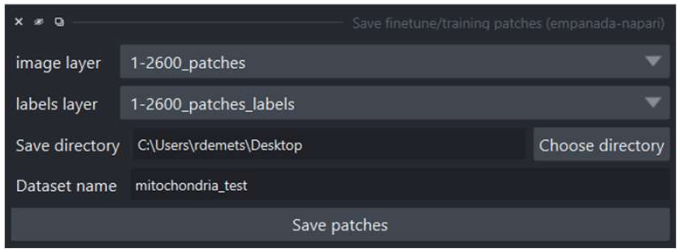
Create training patches by selecting

Plugins > Empanada > Pick finetune/training patches

Delete stack and point layers (keep only patches)
Select the patches_labels layer
Using the brush, draw all mitochondria, starting at label 1
Increment label by 1 at each mitochondria
Save patches

Plugins > Empanada > Save finetune/training patches





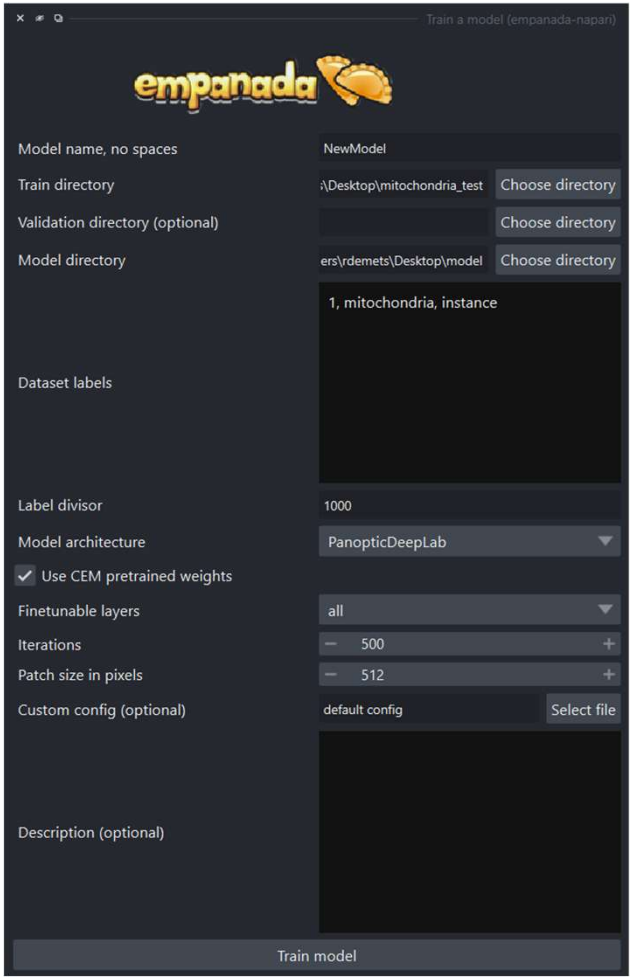
Finetune model **Plugins > Empanada > Finetune a model**



Finetune model	Train new model
Less memory, faster	Computationally heavy
New data close to original data	Need more annotated data
Less accurate	More accurate

Tips
- Annotate all objects in the FOV
- Make patches without objects

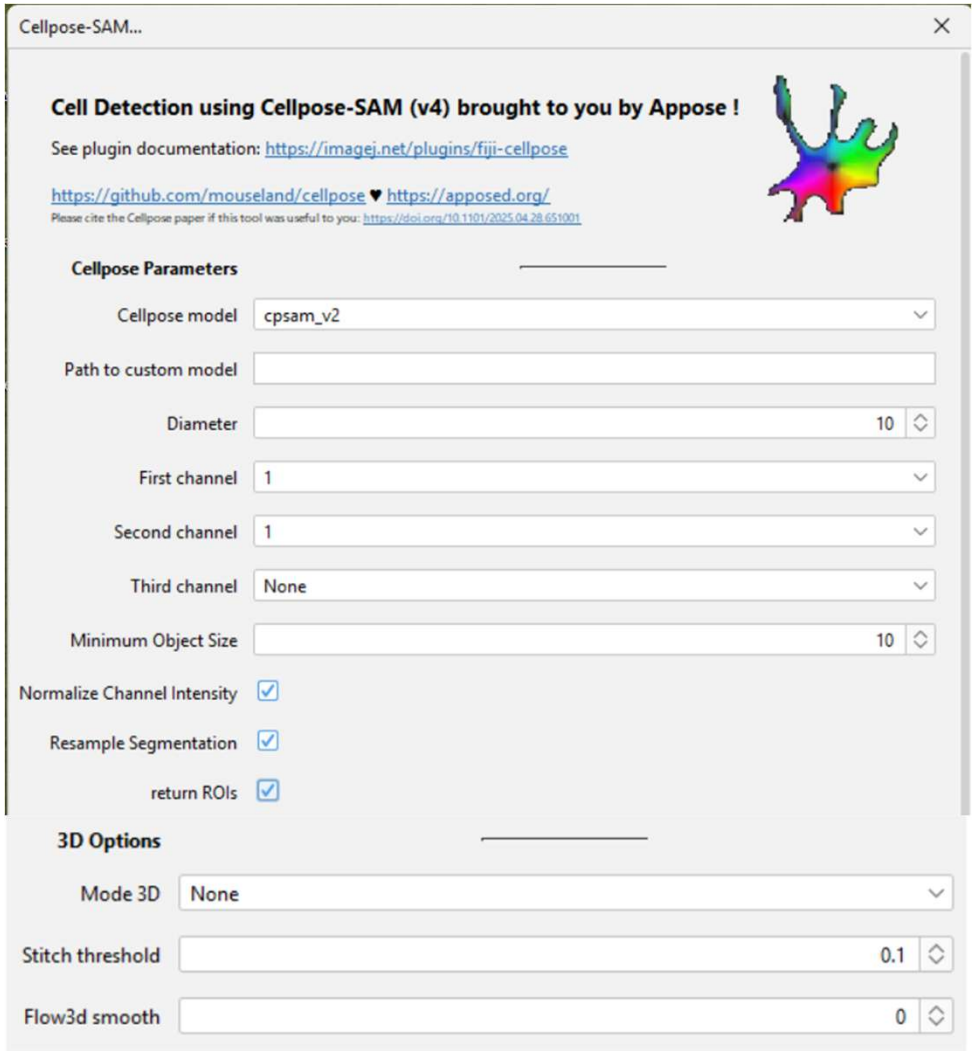
Train model **Plugins > Empanada > Train a model**



Questions ?

2D Test and environment setup

Open nuclei_DSB_2018.tif
Start Fiji > Plugin > Segmentation > Cellpose-SAM
Change model to cpsam_v2
Make sure Mode 3D is set to “None”
Leave all the rest to default and click on Run



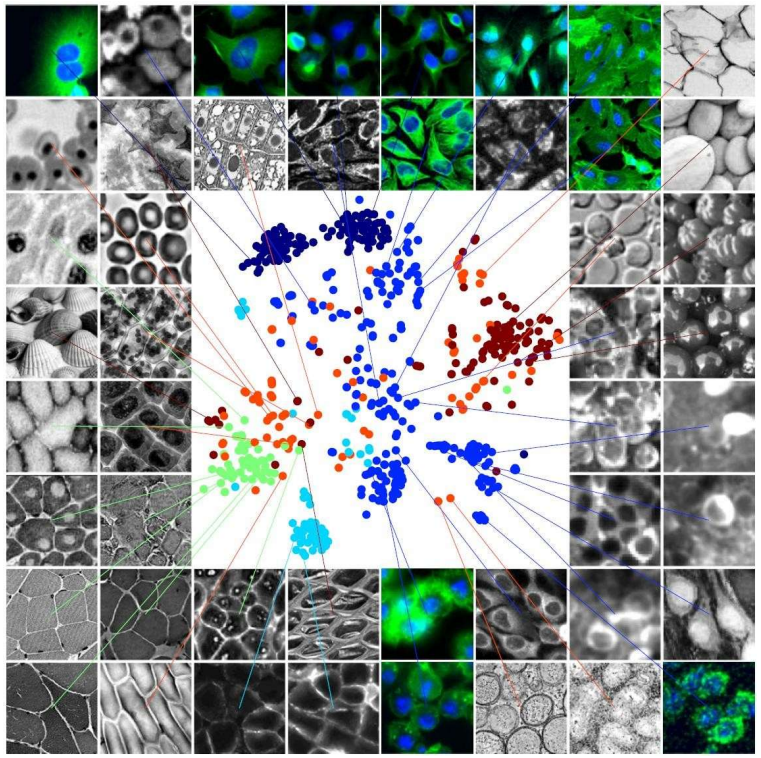
Cellpose version 1 dataset

608 annotated images

Dataset includes:

- Fluorescence images
- Brightfield microscopy images
- Other microscopy images
- Non-microscopy images

→ Better generalization to very diverse cell types



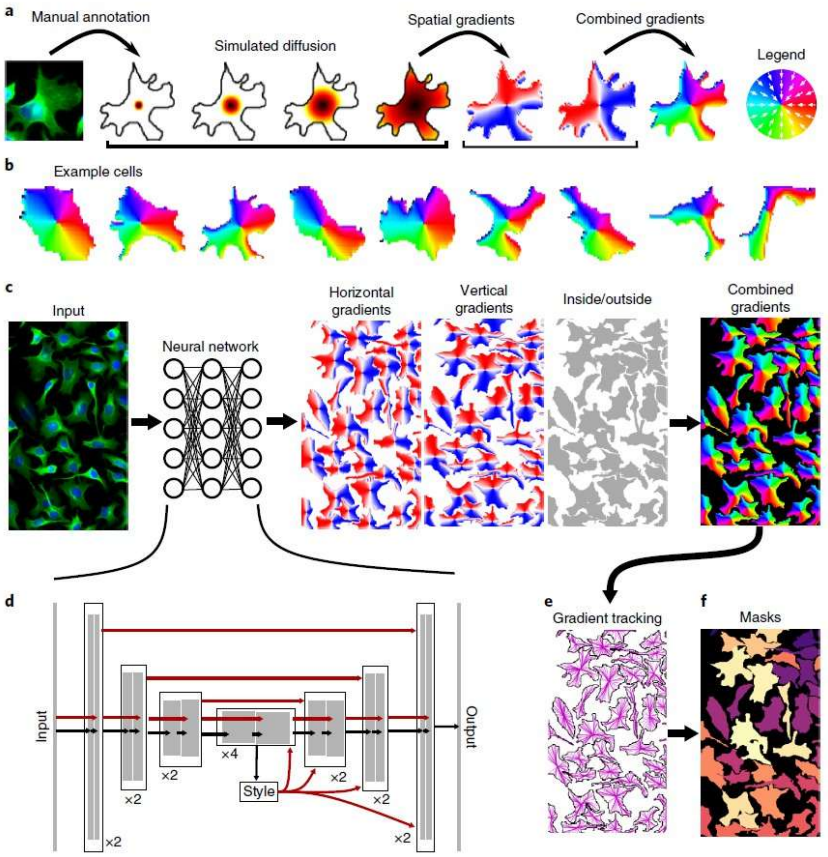
Cellpose – How does it work

Predict:

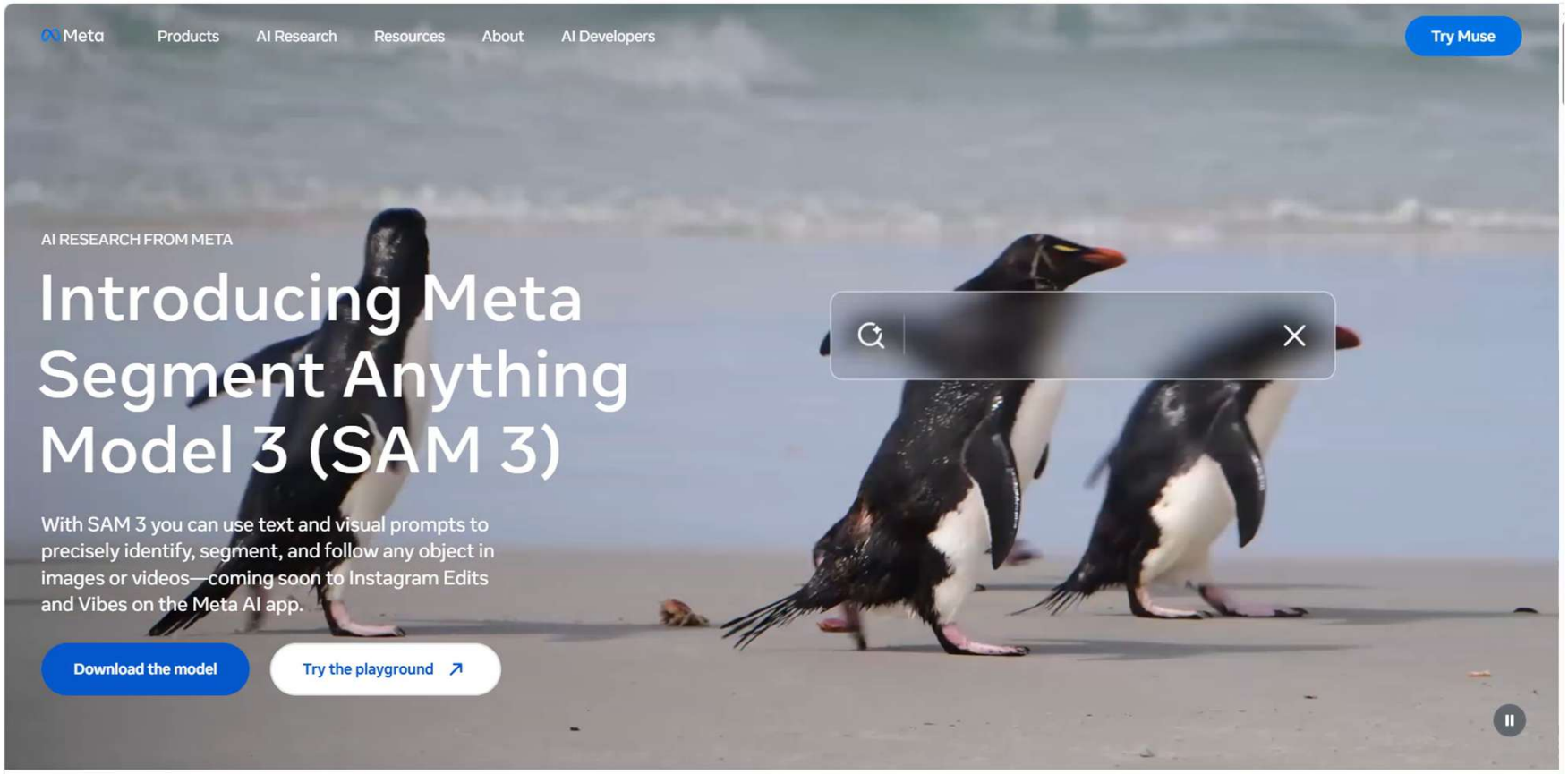
1. Pixel Mask: Which pixels belong to a cell?
2. Two images: (x,y) coordinates of vector pointing to the center of the cell

Find:

All pixels in a mask where the vector-flow ends in the same center, belong to the same cell

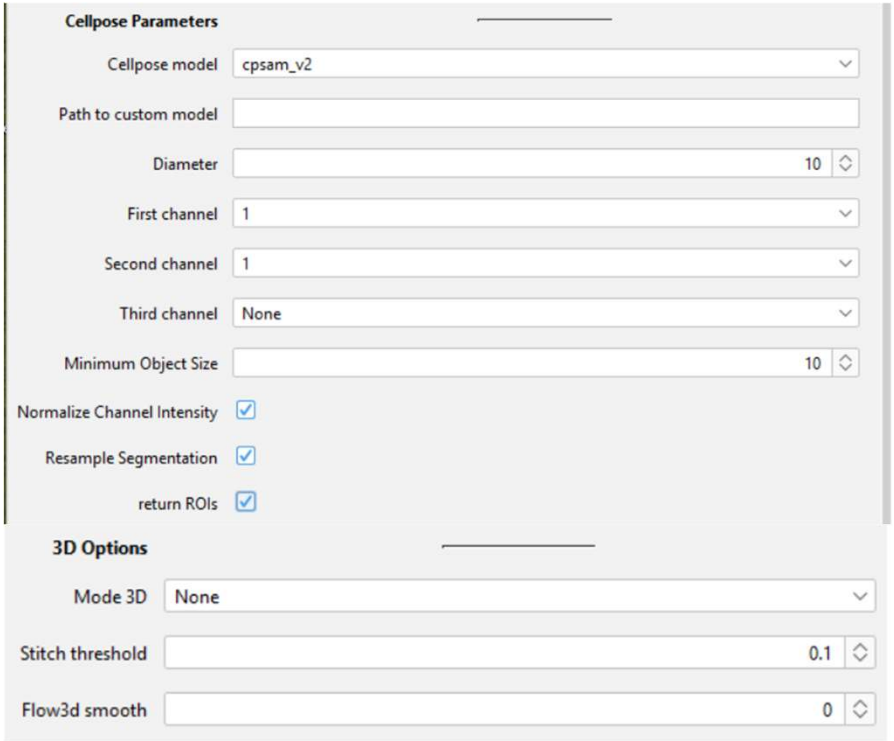


Segment Anything by Meta (Facebook)



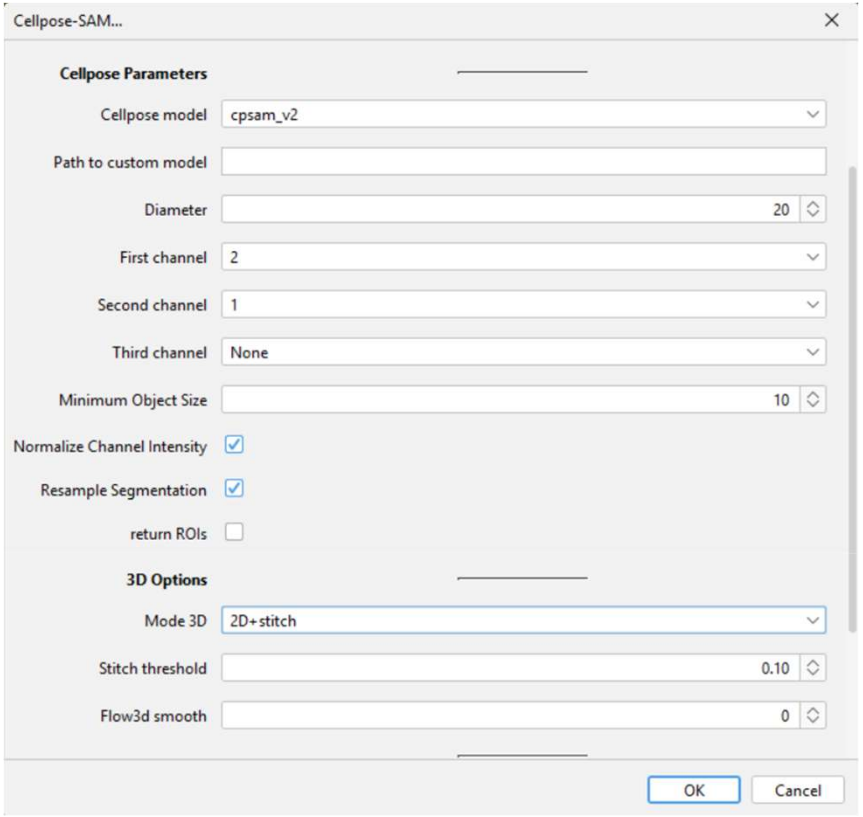
2D Test and environment setup

Open nuclei_DSB_2018.tif
Start Fiji > Plugin > Segmentation > Cellpose-SAM

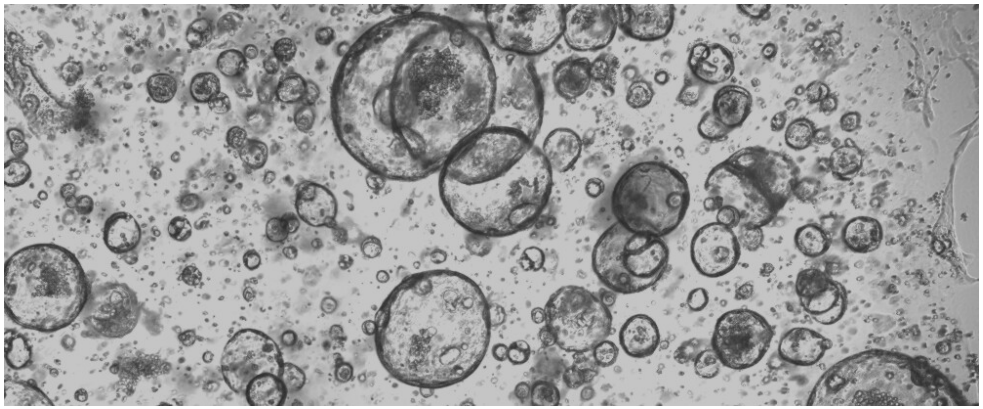


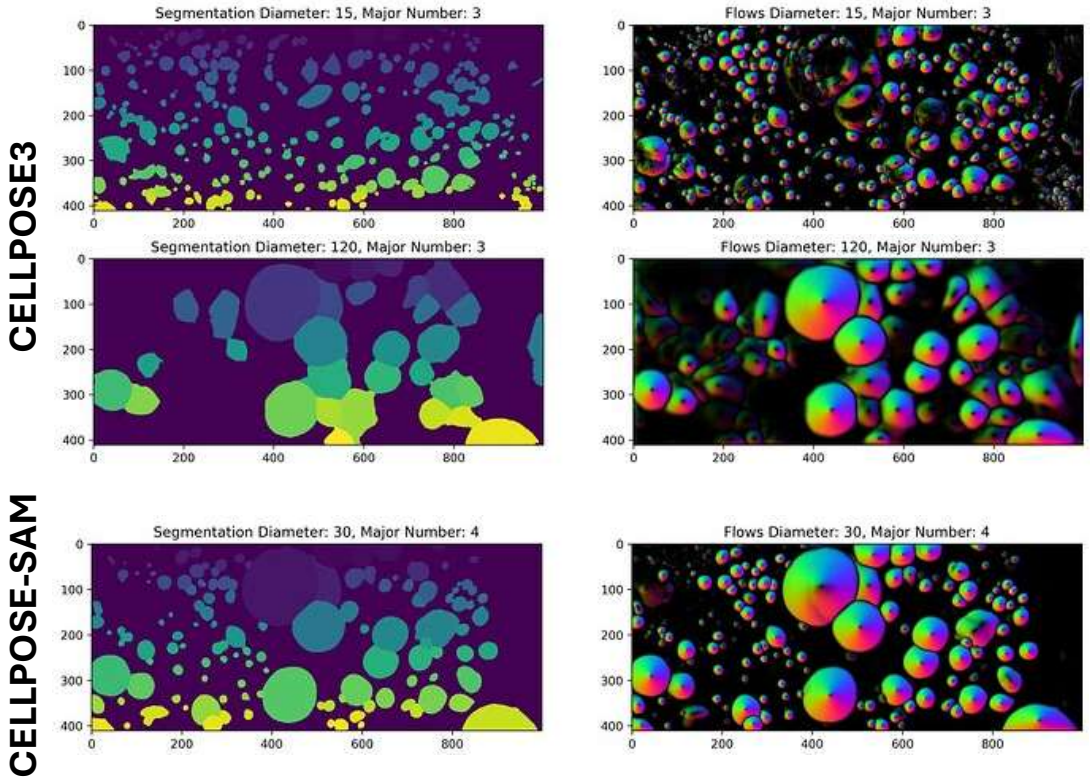
3D Test

Open Cellpose_3D_2ch
Start Fiji > Plugin > Segmentation > Cellpose-SAM



Cellpose 3	Cellpose-SAM
Faster, Size dependency	Slooooow (20x), size independent
Poor 3D capabilities	Very good 3D segmentation
Stable	In development/improvement



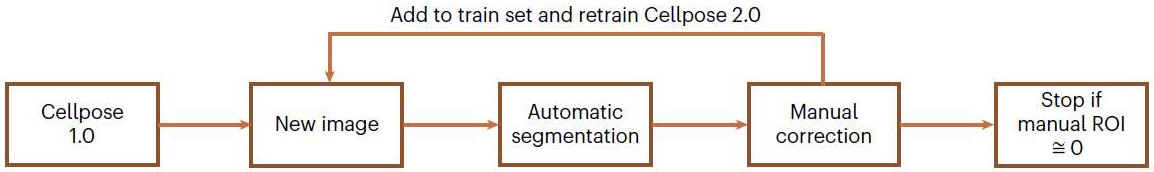


Mode 3D:

2D + stitch : 2D segmentation + Connected Components (Fast)
3D mode : Intersection of XY segmentation + XZ/YZ segmentation (Slow)

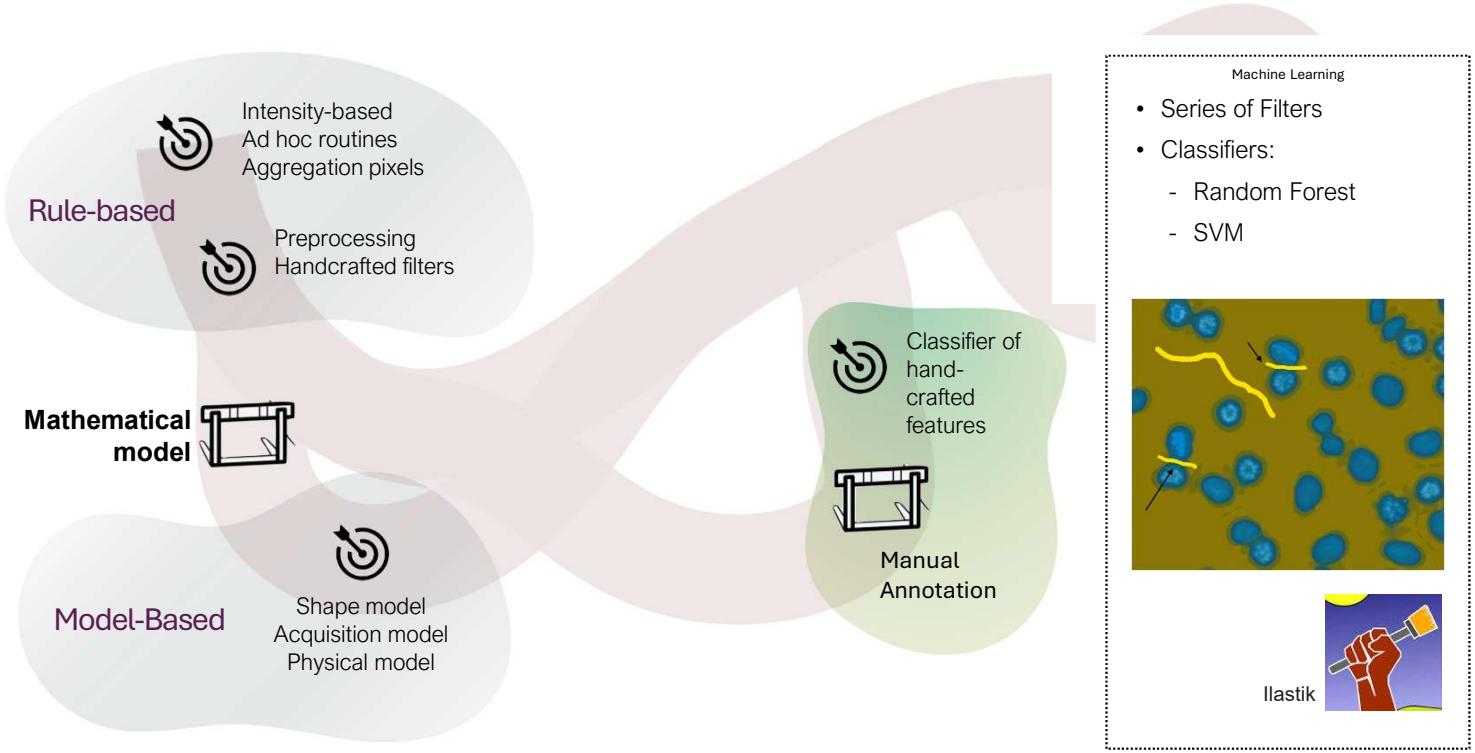
https://forum.image.sc/t/cellpose4-cellpose-sam-tests/

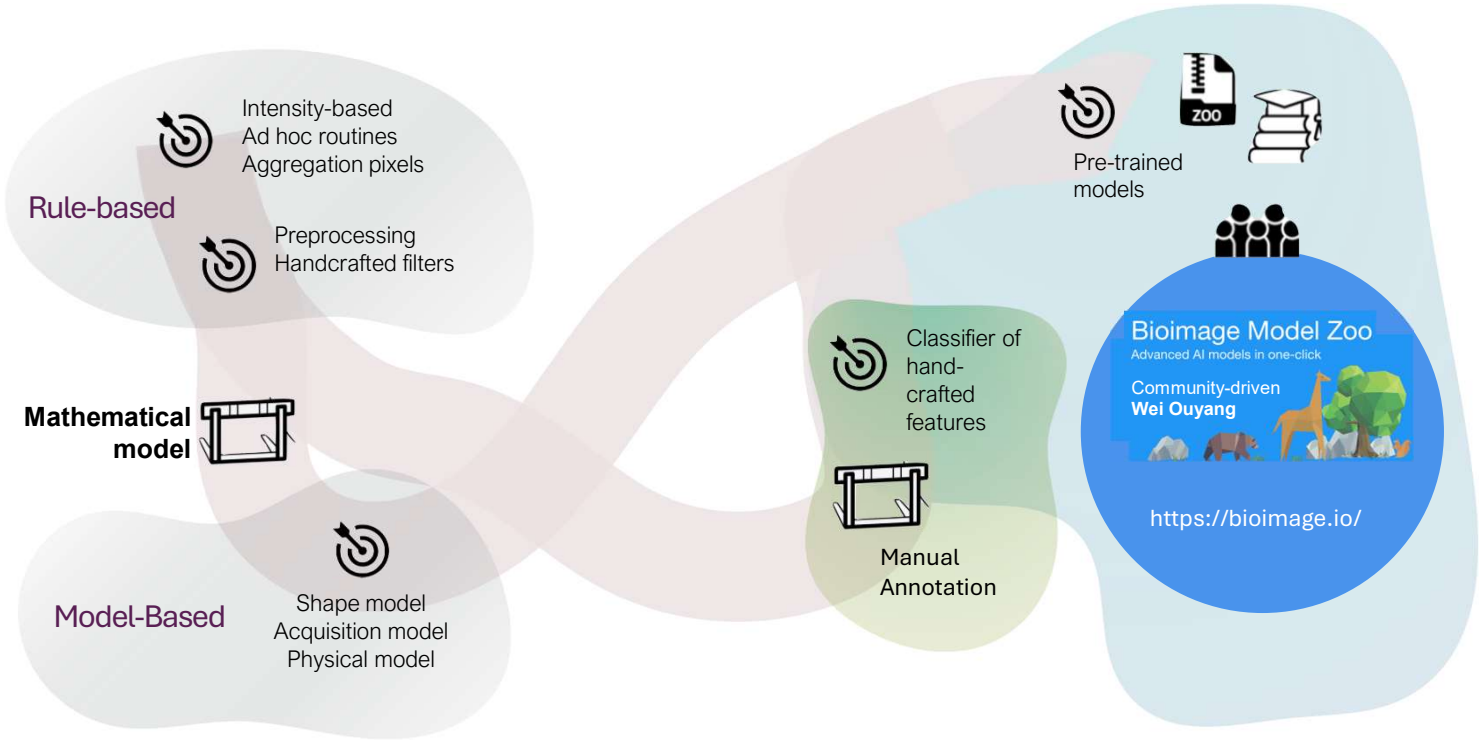
Finetune Cellpose-SAM

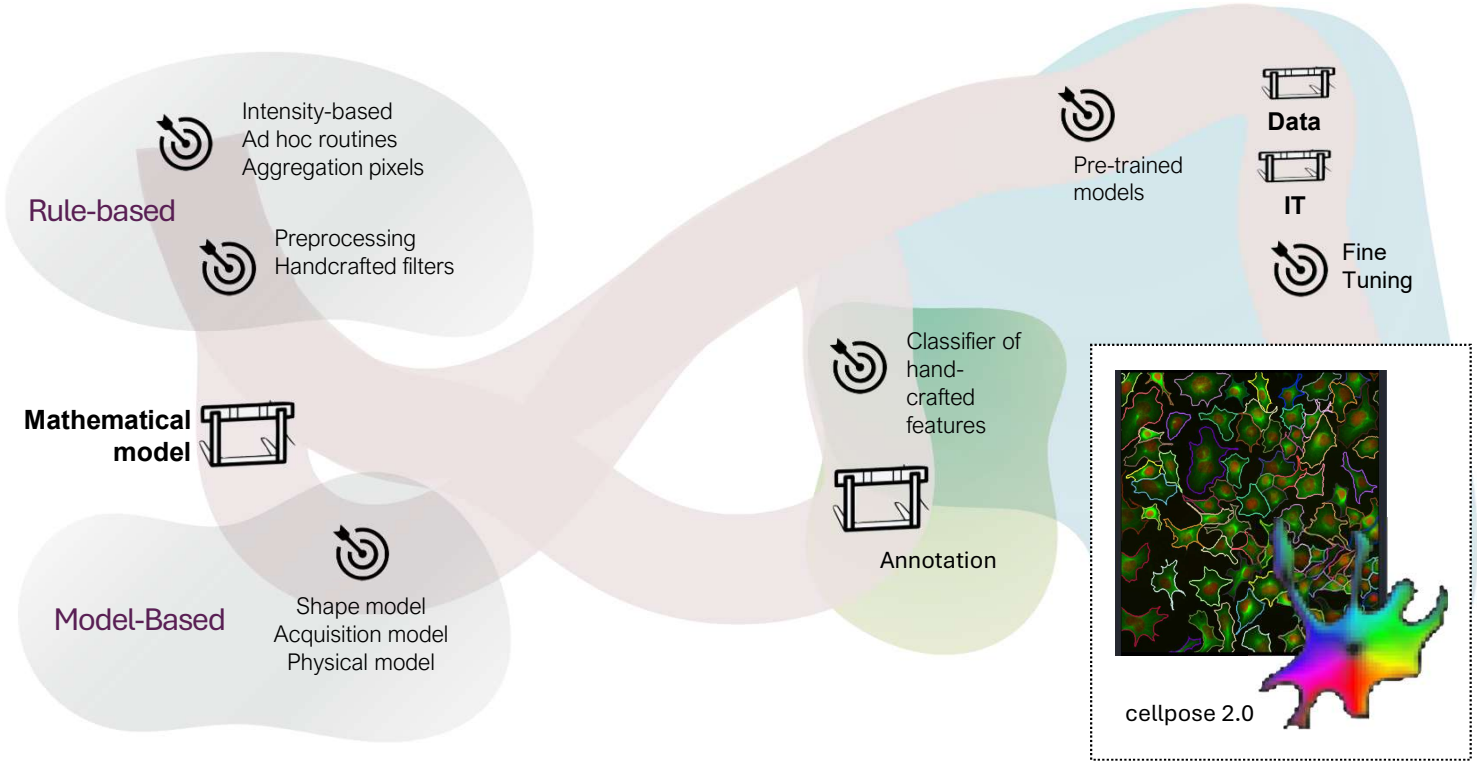


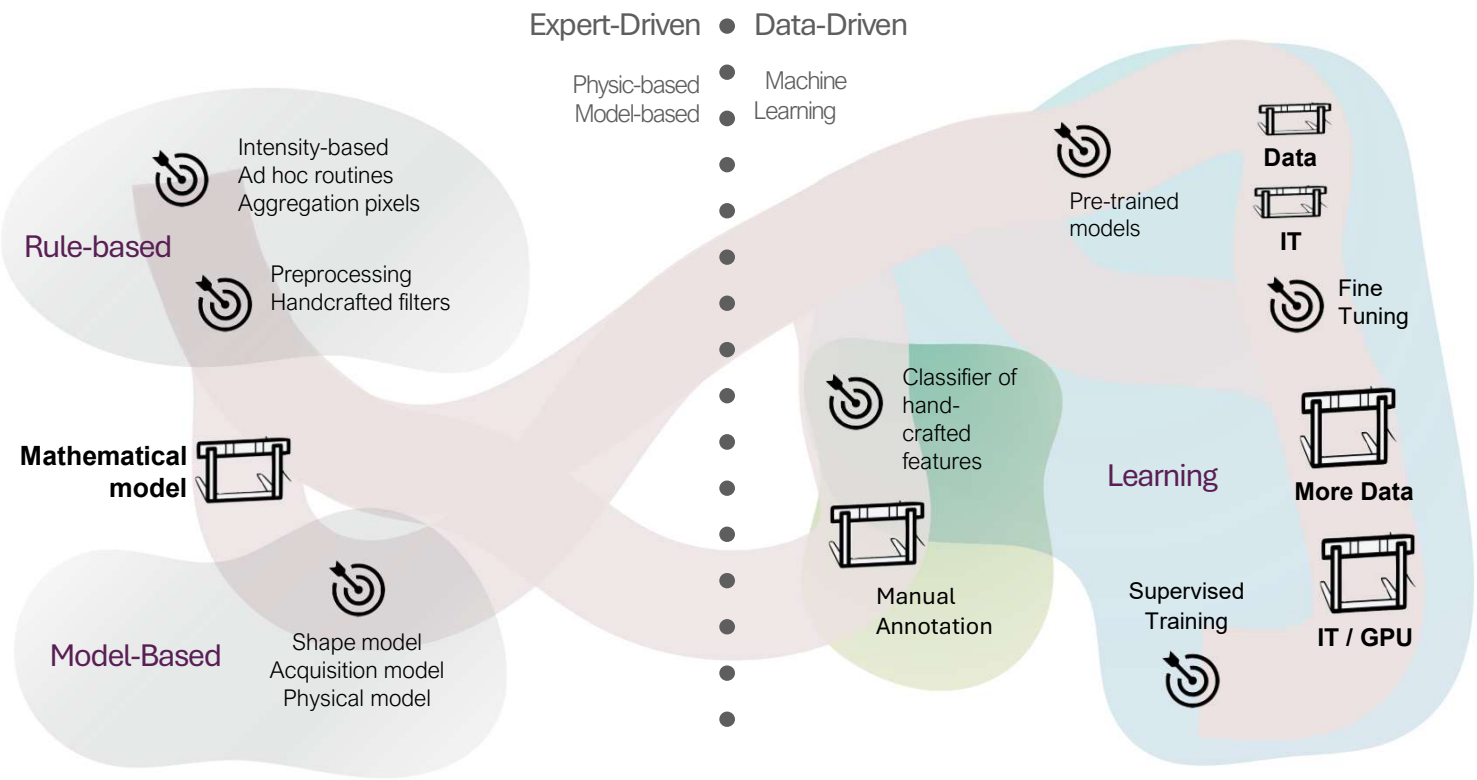
1. Activate env-napari in Anaconda/Miniconda and type “cellpose” to start the GUI.
2. Drag and drop one image from the folder Training>Cellpose
3. Manually curate the segmentation masks. Editing the masks will automatically save a npy file alongside the original image **Drawing**
   - Delete mask: Select with left-click, then press DELETE, or hold CTRL & left-click on the mask you want to delete
   - Draw new mask: Right-click, draw freely, complete mask by moving back to red circle
   - Note: Masks & Outlines can be turned on & off with X / Z. Brush size can be adapted under
4. Repeat on each image of the folder
5. Finetune Model. Open one image from the folder with all redrawn images
   - Go to „Models“ -> „Train new model“ (CTRL + T)
   - Change model name. The default training parameters work well for most cases. Make sure all the images are found for training
   - Press „ok“. The training progress is shown in the terminal
6. After training, the model can be selected under „custom models“ (both GUI and Fiji)
7. Optionally repeat 2 – 6 for more images using the newly finetuned model (more data → more accurate model)

Questions ?


Intensity-based
Ad hoc routines
Aggregation pixels
Rule-based
Preprocessing
Handcrafted filters
Mathematical model
Shape model
Acquisition model
Physical model
Model-Based
Classifier of hand-crafted features
Manual Annotation
Machine Learning
• Series of Filters
• Classifiers:
- Random Forest
- SVM
Ilastik


Intensity-based
Ad hoc routines
Aggregation pixels
Rule-based
Preprocessing
Handcrafted filters
Mathematical model
Model-Based
Shape model
Acquisition model
Physical model
Classifier of hand-crafted features
Manual Annotation
Pre-trained models
ZOO
Bioimage Model Zoo
Advanced AI models in one-click
Community-driven
Wei Ouyang
https://bioimage.io/


Intensity-based
Ad hoc routines
Aggregation pixels
Rule-based
Preprocessing
Handcrafted filters
Mathematical model
Model-Based
Shape model
Acquisition model
Physical model
Pre-trained models
Data
IT
Fine Tuning
Classifier of hand-crafted features
Annotation
cellpose 2.0


Expert-Driven
Data-Driven
Physic-based
Model-based
Machine
Learning
Intensity-based
Ad hoc routines
Aggregation pixels
Rule-based
Preprocessing
Handcrafted filters
Mathematical
model
Model-Based
Shape model
Acquisition model
Physical model
Classifier of
hand-
crafted
features
Manual
Annotation
Pre-trained
models
Data
IT
Fine
Tuning
Learning
More Data
Supervised
Training
IT / GPU

Expert-Driven ● Data-Driven

Model-based		Machine Learning
Based on a-priori model		**Based on data**
Physical rule biophysics law, prior knowledge of the sample	Physics vs. Data	**Adaptivity to the data**
Insight of the structure of the problem Convergence prove	Process Control	**Blackbox** no guarantees, validation on data
Results explainable Mathematics, error analysis	Explainability	**Not explainable** Working in progress, tools
Generalization Under known assumptions	Generalizability	**Strong claim of IA** Questionnable aspect
Very efficient ... Very inefficient from simple algorithms to iterative algorithms	Computing	Training: slow / GPU Prediction: efficient

Carola Bibiane-Schönlieb, ISBI 2021

Has *Classical* IA become useless?

	Classical IA	Deep Learning
Hand annotation required	No	**Yes** (supervised classification)
Specific hardware required	No	Yes (training)
Speed	Fast	**Slow** - Moderate (prediction)
Interpretability	Easy	Hard
Re-usability	Moderate	Low - High (retraining)
Accuracy	Moderate - Good	Best (when trained on sample)
Architecture versatility	Low	High

HOW TO IMPROVE MANUAL ANNOTATION ?